# NOETHERIAN APPROXIMATION PRESERVING GEOMETRIC REDUCEDNESS OF THE FIBRES

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 9844-9851 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9853-9877, the complete author proof (23 body lines).

*Editorial note.* Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Noetherian approximation for finitely presented morphisms of affine schemes and geometric reducedness of fibres; the descent is retained.

*Source reference.* Original lemma 05FB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9739–9753. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0576: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7161–7169. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0579: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7259–7266. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 05F4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9877–9877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to S$ be a morphism of affine schemes, which is of finite presentation with geometrically reduced fibres. Then there exists a diagram as in Lemma 05FB such that in addition $f_0$ has geometrically reduced fibres.*

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* Apply Lemma 05FB to get a cartesian diagram

$$\begin{array}{ccc} X_0 & \xleftarrow{g} & X \\ {\scriptstyle f_0}\downarrow & & \downarrow{\scriptstyle f} \\ S_0 & \xleftarrow{h} & S \end{array}$$

of affine schemes with $X_0 \to S_0$ a finite type morphism of schemes of finite type over $\mathbf{Z}$. By Lemma 0579 the set $E \subset S_0$ of points where the fibre of $f_0$ is geometrically reduced is a constructible subset. By Lemma 0576 we have $h(S) \subset E$. Write $S_0 = \mathrm{Spec}(A_0)$ and $S = \mathrm{Spec}(A)$. Write $A = \mathrm{colim}_i A_i$ as a direct colimit of finite type $A_0$-algebras. By Limits, Lemma 05F4 we see that $\mathrm{Spec}(A_i) \to S_0$ has image contained in $E$ for some $i$. After replacing $S_0$ by $\mathrm{Spec}(A_i)$ and $X_0$ by $X_0 \times_{S_0} \mathrm{Spec}(A_i)$ we see that all fibres of $f_0$ are geometrically reduced. $\square$